

Second part: Open book

Total score: 10 pts.

You are warmly invited to be tidy and concise. Messy papers will be penalized.

6. A political influencer wants to manipulate the public opinion by posting a series of fake news on the social media. The influencer has prepared q different versions of the same fake news and has n fake profiles on the social media where the fake news can be posted. The influencer has to decide how to assign the versions of the fake news to the profiles, so that each profile posts at most one version. The objective is to affect the maximum number of users. The set N denotes the users of the network. Each user $h \in N$ is connected to the set $F(h)$ of fake profiles. A user h is affected if he/she reads the fake news (any version) in at least k fake profiles in $F(h)$. Formulate the problem with a linear optimization model explaining the meaning of the variables and of the constraints. Pay particular attention to the constraints logically linking different sets of variables.

Variante: In order to be more effective, the influencer observed that a user is more convinced of the likelihood of the fake news if he/she sees that at least k **different versions** of the fake news are posted by the fake profiles $F(h)$. Maintaining the constraint of posting at most one version per profile, how does the formulation change?

7. A turbogas power plant has a nominal daily production of b MWh. However, the efficiency linearly reduces with the time and for each day since the start of the plant the nominal production is reduced by a MWh. This means that at day t since the beginning, the production is $b - at$ MWh. From time to time the manager of the plant can decide to clean the turbine to recondition the plant and reset the efficiency of to the initial values. The cleaning operation takes 1 day, and in this period the production is 0 MWh. The price for selling the energy in day i is p_i per MWh. The planning horizon is n days. Formulate the problem of deciding when to do the cleaning so as to maximize the revenue of the plant as the search of a longest/shortest path on a suitable graph. Describe how to construct the graph, the meaning of the arcs and their lengths.